

Data Science Project

1. Data Understanding and Exploration

1.1 Meaning and Type of Features

The Autotrader car dataset contains 402,005 (rows) car adverts with 12 features (columns). Each row represents a single checklist listing. These features fall into 3 classifications: quantitative, categorical and Identifier.

Data type and classifications	Features	Characteristic and role
Numerical (Quantitative)	Mileage, Price, year_of_registration	These features are measured as either continuous or discrete quantities (e.g., the number of car seats), and they are used to apply descriptive statistics for detecting outliers and assessing skewness in the data.
Categorical (Qualitative)	Standard_colour, Standard_make, Standard_model, Vehicle_condition, body_type, fuel_type	These features are descriptive attributes used to classify data. These features are stored as object (string) types and need to be encoded into a numerical format before being used in machine learning models.
Categorical (Binary)	crossover_car_and_van	This feature is a qualitative, binary variable identifying a specific type of vehicle (van or car).
Identifier/code	reg_code, public_reference	This is structural data used for unique referencing. These features can be removed.

Table 1: Data Types and Feature Classifications

The Vehicle_condition feature is a categorical variable, typically stored as an object data type, but can be converted into binary as feature engineering for modelling. This feature is used to classify the car based on its ownership history, which is an important proxy for the car's general condition and provenance. For example, in the dataset, 7.8% of the vehicles are classified as new, while 92.2% are used; this denotes class imbalance. This feature is important, as it can help capture vital relationships; for instance, new vehicles tend to have lower mileage and higher prices, while used cars have greater variability. However, the imbalance in class distribution requires careful attention during the modelling process and training to avoid biased predictions and insights.

The price feature is a continuous numerical (Quantitative) variable that represents the selling prices of the vehicles. This is the feature that will be used to do the prediction modelling task, for example, estimating market value, identifying pricing anomalies and generating insight into depreciation trends. The feature displays a wide range of prices, beginning from £120 to £9,999,999; this clearly indicates extreme outliers. To ensure more accurate insights and predictions, a winsorized version of the price was created, cutting the extreme values while keeping the importance of the overall structure of the data. This method price distribution of this method has a mean of £16,435.7 and standard deviation of £14,463.7, with the median being stable at £12,600. The role and characteristics make the price a central and informative feature for understanding vehicle valuation across different market segments.

The body_type feature is a categorical (Qualitative) variable, used to classify vehicles based on their design and proposed use. It includes multiple groups such a saloon,

Estate, Coupe, Hatchback and SUV, and many others. The dataset is dominated by hatchback (41.8%) and SUVs (28.8%) compared to most of the body types in the dataset, implying a high popularity for this design(styles). There is an imbalance, and it may influence model performances for in classification task or when analysing price trends across body_type. Understanding the categories is important for segmenting the markets and identifying patterns in vehicle cost, values, customer presence and performance.

1.2 Analysis of Distributions

The bar plot (Figure 1) of the percentage of missing values shows several features that contain missing data. Most missing data is concentrated in a few features. year_of_registration (8.29%) and reg_code (7.92%) are the most missing data, indicating that these two features will require the most attention during the data cleaning. The feature standard_colour also shows a large percentage of missing data at 1.34%. Compared to features like body_type (0.21%), fuel_type (0.15%), and mileage (0.03%) are almost complete. The bar plot highlights a focused data quality issue rather than a widespread data collection problem, suggesting that an imputation or exclusion plan should be planned for the incomplete features.

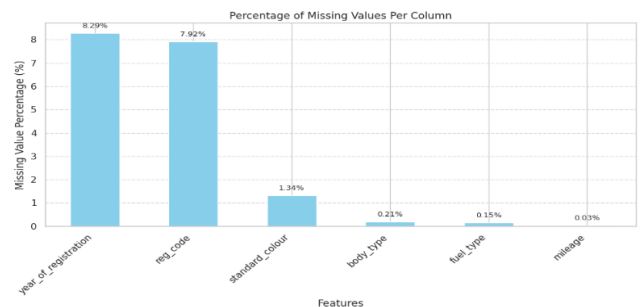


Figure 1: Bar plot showing percentage of missing values.

The histplot (Figure 2) of price features is highly right-skewed (positively skewed). Most of the vehicles are priced lower, with the peak frequency(mode) happening between the 5,000 and 15,000 price range, but extreme upper values stretch the scale. Figure 2 clearly suggests the presence of these outliers, and it's recommended that further analysis is necessary, as we have a couple of expensive cars in general; for example, Ferrari, Lamborghini, Porsche, Bugatti, McLaren, and Aston Martin can cost up to 3 million. However, seeing 9999 million costs on a used car is an outlier. It's important that data cleaning be carried out here because the large outliers can disproportionately influence statistical calculations on the performance of the model and insights; instead, by removing them, the data are transformed to handle these outliers.

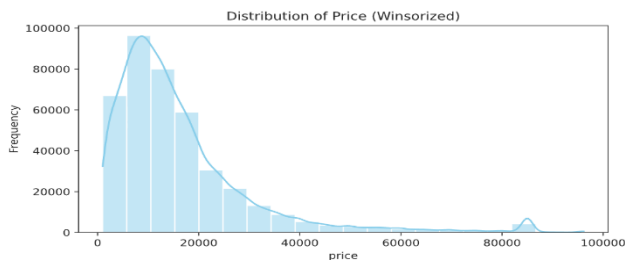


Figure 2: Histogram showing the price column is heavily skewed to the right.

The boxplot (Figure 3) showing the distribution of mileage, including the median, quartiles and outliers, reveals a classic pattern for usage data, which is highly right-skewed (positively skewed). Most of the cars have low mileage, with a median of 35,00 miles. 50% of the data is in the range of 15,000 miles(Q1) to 55,000 miles(Q3). However, the figure shows a massive number of outliers, representing the cars with extremely high mileage values. This suggests data cleaning is required, so it ensures the feature does not disproportionately bias the predictive model.

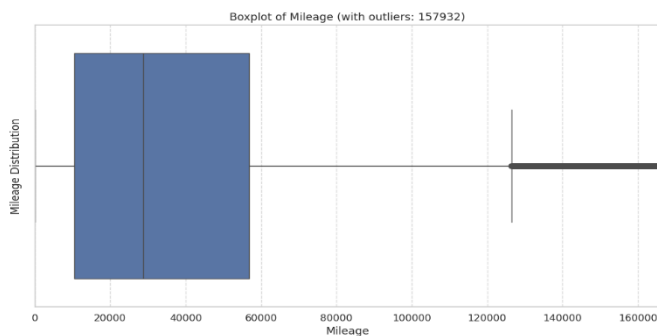


Figure 3: A box plot showing the distribution of mileage, including the median, quartiles and outliers.

The boxplot (Figure 4) showing Year of Registration by price bucket illustrates a strong correlation between price bucket and the vehicle's year of registration. For example, budget-friendly cars show the largest spread, with registration years ranging from 2000 to 2020. The median year is 2015, suggesting that the cheapest cars are older. There are many falling well before 2005, which the outliers confirm that there are older models. Cars in the mid-range price stick closer together, showing tight distribution

clustering near 2019, suggesting more consistency and less variety. Luxury and high-performance cars lean strongly toward more recent years, with a median of 2020 and showing fewer outliers, underlying their premium position and newer manufacturing trend. This boxplot captures the average, spread and anomalies clearly, revealing subtle links between the price and the year of manufacturing while proving actionable insights for market segmentation and planning.

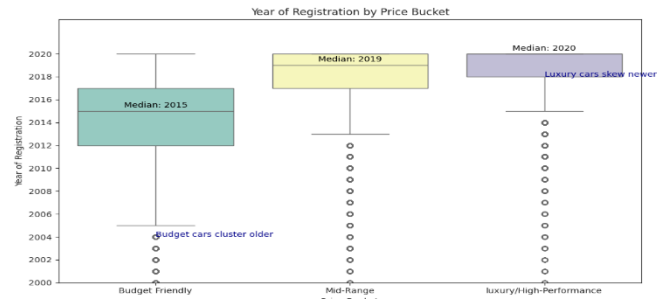


Figure 4: Box plot showing Year of Registration by price bucket.

This violin plot (Figure 5) displays the cost spread among the leading 10 regular car brands, illustrating how variability and central tendency differ within the dataset. Instead of just bars, a violin plot was used. The chart highlights clear gaps in pricing spread and density among other brands. For example, Land Rover offers the largest range and has the most cars priced above £100,000, which points to it being in the luxury segment. Mercedes-Benz, BMW, and Audi carry widespread distribution with dense Mid to high price clusters, indicating their positioning as high-performance brands. However, Vauxhall, Peugeot, Toyota, Nissan and Renault show narrower distributions mostly down low on the price range, implying these brands are cheaper and have market accessibility as they reach more buyers. Figure 5 clearly shows how the data spreads up and down and highlights outliers and uneven patterns where applicable. The analysis provides a deep understanding of how brands set prices and split up the market, which helps figure out where companies stand against competitors and who's most likely to be busy.

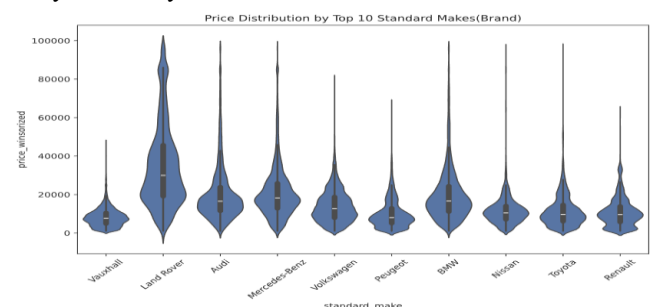


Figure 5: Violin plot showing Price Distribution by Category (Brand).

2. Data Pre-Processing

2.1 Data Cleaning

There are 33,311 missing values in the year_of_registration feature. To address this large gap while avoiding the significant data loss associated with dropping rows, an imputation approach was implemented. The first step taken

was filling the missing `year_of_registration` based on certain conditions, for example, if the mileage is greater than or equal to 0 and less than or equal to 2000, and the `reg_code` is missing or empty, and the vehicle condition is equal to “NEW”, replace missing values in `year_of_registration` with 2020. This is done because the dataset is 5 years old and the max year is 2020; thus, any new record on the dataset can have 2020 as the year of registration. However, it improved the data quality.

Another similar logical imputation method was used for inferring missing registration years for “USED” cars from partial `reg_code` records. First, a new temporary numerical column of `reg_code` was created to identify records that can be converted. For the ones that were converted, then the code fills the missing `year_of_registration` if the `reg_code` is between 2-9, mapped to `year_of_registration` 2002-2009 and between 10-19 to years 2010-2019, by adding 2000 to the numeric `reg_code`. After filling in the missing values, the temporary column was removed to maintain data quality. After implementing these methods, there were still significant missing values for the `year_of_registration` and `reg_code` columns. To address this and enhance the accuracy of the incomplete records, a detailed mapping dictionary (`reg_to_year`) was created, linking UK vehicle registration codes (e.g., 51, 02, 70) to their corresponding year of registration.

Before using the mapping, the `year_of_registration` column was converted to a nullable integer type to adjust the missing values, and the `reg_code` column was cleaned by converting it to a string and removing any whitespace. The mapping was then used to fill in the missing `year_of_registration` entries where the `reg_code` matched DVLA registration patterns. For example, vehicles with `reg_code` “09” or “59” were assigned a `year_of_registration` of 2009, while those with “20” or “70” were assigned 2020. This method was chosen because it aligns the data with actual registration identifiers, ensuring accuracy and consistency for analyses, such as tracking price trends over vehicle age.

To manage missing values in the standard `colour` column, a targeted imputation approach was used. First, the average colour for each brand was calculated and applied to fill missing values within that group. If the standard make had no average colour, the most common colour in the dataset was used instead. This imputation technique ensures that missing values are replaced with contextually relevant data, enhancing the consistency and accuracy of the model's predictions and insights.

However, similar imputation were handled missing values in the mileage column, using a grouped median approach. The dataset was grouped by standard make, standard model and `year_of_registration`; these are features likely to influence mileage. With each group, the median mileage was calculated and used to fill missing values, and if the group lacked sufficient data, the overall median mileage of the dataset was used as a fallback. Mileage is an important factor in pricing and buyer decision-making; if the column is missing values/data, it may be undervalued or excluded

from the pricing model. Thus, by imputing missing mileage using the grouped median, maintain pricing accuracy.

These methods improved the completeness, reduced noise and prepared the dataset for featuring engineering, categories comparison and insights.

Handling outliers in price: Figure 6 shows price feature had an extreme data quality issue caused by the extreme positive outliers (up to £9,999,999 and heavy right-skewed skewness, which an inaccurate value is £17,343 and a standard deviation of £46,437. To handle these outliers without losing valuable information and balancing between minimising the influence of extreme values, winsorization was selected over other methods, such as removing and replacing. The method was applied to the 1st and 99th percentiles and, importantly, was stratified across the vehicle_condition categories (New and Used). This stratification ensured outliers were capped relative to their own market segment, and the resulting summary confirmed the success: 1.99% of new cars and 1.95% of used cars were capped, guaranteeing neutral and equitable treatment across both segments. Making sure there are neutrals and reasonable treatment across both segments. After the winsorization, the maximum price decreased from £9,999,999 to £ 96,201, and the standard deviation to £14,463, while the median and interquartile range remained stable. The final distribution still shows a long upper whisker above 40,000; this means some high values are still far from the median but are no longer true outliers in a statistical sense. That’s why 0.01% was used in the Winsorization. This preserves realistic high-end price data relevant to luxury vehicles while effectively removing inaccurate extreme noise, therefore offering more stable and reliable central tendency without loss of statistical power.

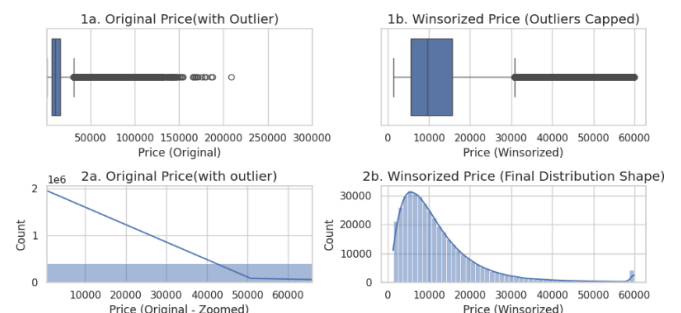


Figure 6: box plot for outlier and range, and histograms for shape and skewness comparison.

2.2 Feature Engineering

A new feature called `log_mileage` was engineered from mileage. The mileage distribution was highly positively skewed with large variance, as shown in Figure 6, making the analysis difficult because most details were visually compressed. To address this, a logarithmic transformation was applied using the formula: “`car_dataset['log_mileage']=np.log(car_dataset['mileage_winsorized'] + 1)`”. This feature compresses the high-value tail and expands the lower-value clusters, producing a more balanced distribution. The feature engineering was essential to reduce skewness and variance, improving interpretability

and robustness for subsequent analysis. The log transformation reshaped the distribution to be more symmetrical and centralised, which is beneficial for models that assume normality and for visualising patterns in the data. The left plot shows the original mileage distribution, which is heavily skewed with most values compressed at the lower end. The right plot shows the log-transformed distribution, which is more symmetrical and centralised, reducing the influence of extreme values and improving visualisation.

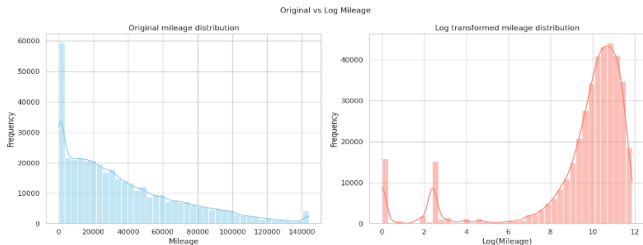


Figure 7: Histogram showing original mileage vs log mileage

A new combined feature named `model_popularity` was created to display how appealing each car model is, as shown in Figure 8. The score pulls together two normalised components: how often a model appears (demand) and the mean price. Both are scaled to a 0-1 range so they can line up fairly. A weighted formula (60% frequency, 40% price) was used to highlight interest more than cost and still include pricing stance. The score was mapped to each value, and popularity classes were created using quantiles based on segmentation and labelled ‘Low’, ‘Medium’ and ‘High’. This feature was created by figuring out how often every model shows up by running `value_count()`, picking the typical cost for each type by grouping data, then figuring out the average, and normalising both metrics to a 1-1 scale using min-max scaling. Put them together using weights, $\text{weighted_score} = (0.6 * \text{model_freq_norm}) + (0.4 * \text{price_norm})$, and finally mapped the score back to the dataset and created popularity classes using `pd.qcut()`, which sorts values by rank instead of fixed ranges. This feature is added because it clearly shows a picture of how appealing cars are in the market, showing what customers like and how price plays out. By combining demand and price, it helps forecast things like future sales worth, stock handling or focused advertisement. Turning numbers into a common scale keeps comparison fair, and splitting them into groups makes results easier to use later. For example, models like Golf, Corsa and C Class are most liked, as they are popular and have competitive pricing, while the distribution classes (Low: 134,154; Medium: 133,977; High: 133,874) show an even spread across segmentation.

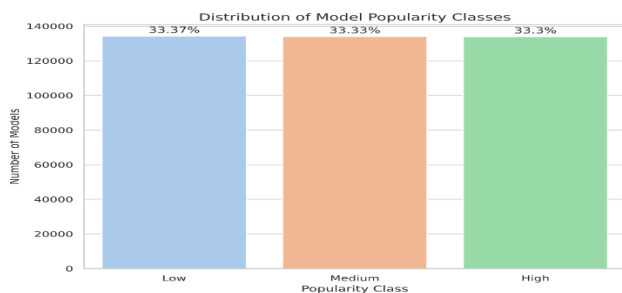


Figure 8: bar plot showing the distribution of model popularity classes.

A categorical feature called `price_bucket` was engineered to group cars into three clear price ranges for easier analysis and market segmentation. Budget-Friendly (below £18,000), Mid-Range (£18,000 up to £50,000), and luxury/high-performance (over £50,000). `Pd.cut()` was used with defined bins and descriptive labels applied to the winsorized price column. This classification adds clarity when comparing vehicle make, model, fuel type, body type and other columns with price and helps visualisations and predictive modelling. The pie (Figure 9) showing the proportion of price buckets displays budget-friendly cars of 281,151 (69.9%), mid-range cars of 105,761 (26.3%) and luxury/high-performance cars of 15,093 (3.8%). It clearly denotes real-world market patterns showing cheaper options dominate while luxury vehicles form a smaller but significant niche. This feature was used as the price column was unbalanced and showing extreme outliers. It is also created to support target pricing strategies, separating customers and tracking trend analysis across different market segments.

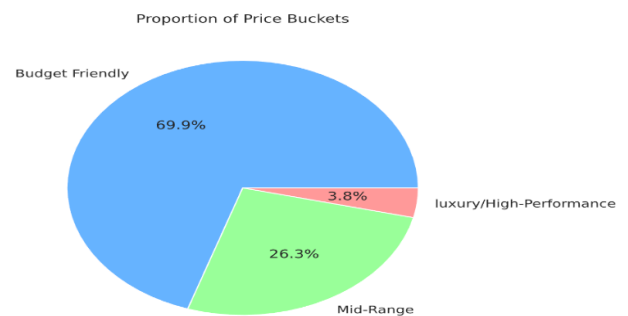


Figure 9: pie chart showing the proportion of price buckets

A new categorical feature called `New_fuel_type` was created to address imbalance and sparsity issues in the original `fuel_type` column, shown in Figure 10. The original dataset contained many rare fuel types such as “Bi Fuel”, “Diesel Plug-in Hybrid”, and “Natural Gas”, which had very low counts. To simplify and balance the data, major categories Petrol and Diesel were preserved, all hybrid variants were combined into “Hybrid”, Electric remained separate, and rare types were grouped under “Other”. The original `fuel_type` feature was imbalanced and sparse, making insights and visualisation difficult. Grouping rare categories improves interpretability, balances the distribution, and supports clearer comparisons across fuel types in predictive modelling. The left countplot shows the original distribution, where petrol and diesel dominate, while other categories have extremely low counts, indicating sparsity. The right countplot shows the mapped distribution after grouping, which is more balanced and interpretable. Petrol and diesel remain major categories, hybrid and electric form smaller but significant segments, and “Other” captures rare types.

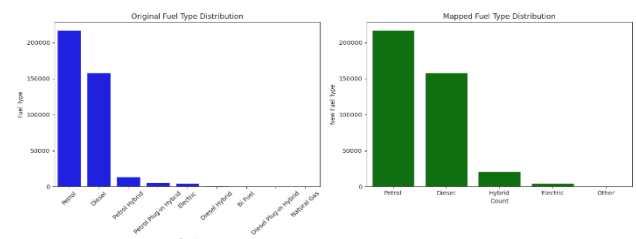


Figure 10: Count plot showing side-by-side fuel type distribution

2.3 Subsetting

According to industry updates and news articles, UK car sales dropped sharply during COVID, with new registrations falling almost 30% in 2020 due to lockdowns disrupting deliveries and eroding buyer trust (Jolly, 2020). However, electric car sales held better slightly in the wider market, down just 10% (Jolly, 2020). Using our dataset, we decided to investigate how this shift influenced used car prices. Following clear filtering rules/subsetting, first focused on used vehicle in 2018 and 2020, then created comparable sets using car brand, model, body type and fuel type to ensure like for like comparisons so it stayed fair, then narrowed mileage between 5k and 30k miles to control for usage and finally for robustness, kept data group with at least 5 or more listing per year while holding its reliability. These steps help spotlight change from registration data alone while maintaining major price factors steady. This method removes confounding factors, for example, mileage, age or model variety, focusing straight on the business question. The business question is whether used car prices increased during COVID. Doing a year-to-year comparison can be misleading as mileage, age and vehicle condition strongly influence price and will be the right features for prediction.

Therefore, the histplot (Figure 11) showing within within-group median price difference helps answer our question. Once the data quality and relevance improvement were made, the output gave us clear insights. Figure 11 shows similar cars in 2020 price increase of about £7,100, with 98% of specs higher, meaning most of the cars were expensive, and the median like-for-like ratio was 1.37, suggesting a 37% increase in 2020. This tells us how prices shifted during the pandemic.

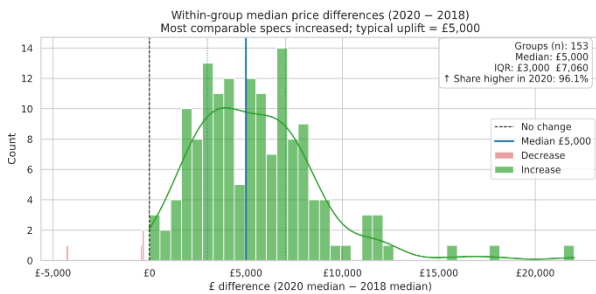


Figure 11: A histplot showing within within-group median price difference,

The second subset focused on year_of_registration from 2015 to 2020 and the four major fuel categories: Petrol, Diesel, Hybrid, and Electric. This was implemented using conditional selection, for example, (year_of_registration >= 2015 and New_fuel_type in the specified list), ensuring transparency, limiting results to recent years and removing.

Irrelevant noises. The purpose was to focus on recent market dynamics and fuel technologies relevant to the current industry trend. Electric and Hybrid vehicles were included to capture adoption growth, while Diesel and Petrol provide a baseline for comparison. This method improves data quality by excluding older cars' data, which might distort findings, and improves relevance by focusing on categories with significant market impact. Normalisation was applied to calculate proportions per year, allowing fair comparison across years with varying registration sections. The line plot shows the normalised fuel type proportion from 2015 to 2020.

This line plot (Figure 12) highlights three key trends: diesel decline; diesel cars decreased from 54% in 2015 to 22% in 2020, likely due to stricter emissions regulations and people choosing different cars. Petrol peak; petrol remained leading fuel used, but the highest point at around 64% in 2018 then settled close to 60%, suggesting temporary market fluctuations. Electric and Hybrid Growth; hybrid climbed up from 4% to 12%, while Electric rose from 1% to 5%, showing steady adoption of low-emission fuels. These findings align with wider moves across the automotive industry, pushing sustainable mobility. The use of normalised values ensures comparison across years, while narrowing focus to the latest data and primary fuel sharpened insights into current market trends.

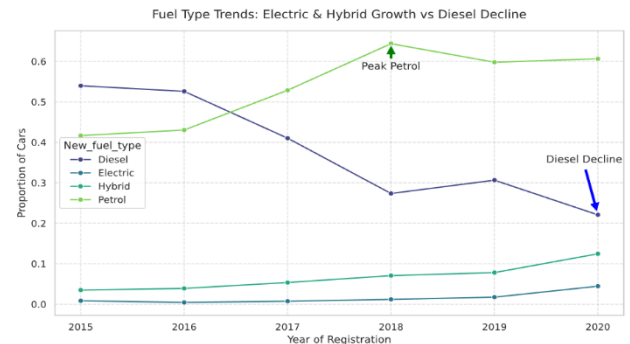


Figure 12: Line plot showing fuel trends for the recent years and relevant fuel types

The third subset focused on comparing prices between low and high popularity models. The data was filtered to separate vehicles with similar mileage and age. Only vehicles aged 3 to 7 years with mileage between 20,000 miles and 80,000 miles were kept, and included only models that had at least 100 cars in each popularity class. This controls for depreciation and removes noise from rare models. It also ensures a balanced and reliable comparison setting. The reason for this subsetting is to investigate whether popularity influences pricing when other major factors are held constant. A boxplot was used to compare prices between the popularity classes. To tackle extreme values and improve the visual readability, the y-axis was transformed to a log scale, making the main price range fill the graph instead of being squeezed at the bottom by very extreme outliers.

Figure 14 shows that high-popularity models have higher median and mean prices compared to low-popularity models, even though the interquartile range is similar for

both. High-popularity models show more outliers, showing that some luxury and high-performance cars exist within this group. Most prices cluster between £10,000 and £50,000, even though expensive cars go much higher. These findings suggest that popularity has a noticeable impact on pricing, mainly in the mid-range market segment.



Figure 14: showing price comparison: low vs high popularity models.

3. Analysis of Associations and Group Differences

3.1 Quantitative – Quantitative

Price and mileage association (depreciation analysis)

According to Autotrader's explanation of depreciation in their news article, vehicles' value tends to decrease over time due to wear and tear and, most significantly, increased mileage (Woodhouse, 2023). Using our dataset, we decided to investigate this important industry market principle by analysing the relationship between car mileage and listing price. Both price and mileage features were already winsorized to reduce the skewing effect of the extreme outliers. Given that our dataset is large, we used a hexbin density plot instead of a standard scatter plot to effectively visualise the concentration of listings and identify the key market trends without overplotting. The analysis proves a strong negative correlation between the price and mileage, which fits how vehicles usually lose value.

The regplot showing the relationship between price and mileage (Figure 13) shows, most cars are found in the low mileage, mid to high price segment. This denotes that Autotrader mostly sell cars that are used with low mileage and newer cars. Most importantly, the depreciation curve is nonlinear. The rate of value loss is from 0 to 20,000 miles. The vehicle loses its value fast, since it's no longer new, but as mileage increases over 80,000 miles, the curve decline slows down a lot, suggesting that the price doesn't fall as much anymore, because the car's usefulness starts to set a price floor. This non-linear trend is vital as it signals that accurate fair market prediction requires models that handle how uneven the depreciation can be.

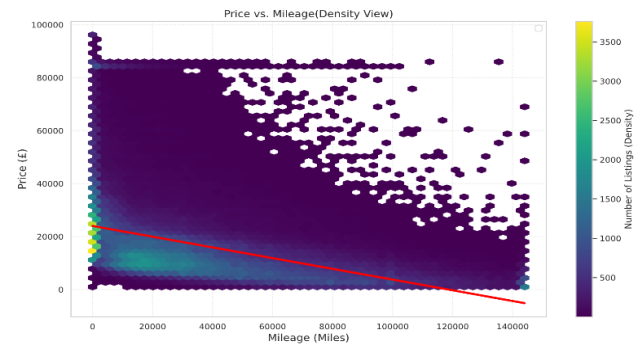


Figure 13: Regplot showing the relationship between price and mileage.

Price and Vehicle Age association (depreciation analysis)

According to Autotrader's the depreciation rate can vary significantly depending on several factors, including its age (Woodhouse, 2023). This means the cars lose value over time, mainly due to age and wear. To validate this with our dataset, we analysed the relationship between vehicle_age and price. The price was already winsorized to reduce the influence of extreme outliers. A hexbin density plot was used rather than a standard scatter plot to visualise effectively and concentrate patterns, trends and avoid overplotting. Figure 15 shows a strong negative correlation between the age of the vehicle and price. The older the cars, the cheaper they become. The depreciation curve is not straight, which means the vehicle loses value faster in the first few years, then much more slowly as it gets older. This trend is necessary for understanding market performance and building precise models.

Based on observation, (figure 15) demonstrate, prices decrease quickly during early years (0-10 years); suggesting rapid depreciation as the vehicle change from "New" to "Used" and warranties expire; depreciation slows after the first decade, because vehicle have already lost most of their values and remaining price reflects how useful it still is (functionality), not how new or trendy it looks. Vehicles that are more than 30 years old, their price stops decreasing and remains almost the same, forming a price floor. However, some very old cars become more expensive, likely because collectors want them as classic cars. Figure 15 denotes that most of the cars fall within the low age, mid to high price segment, suggesting that Autotrader mainly sells newer or a little used vehicles.

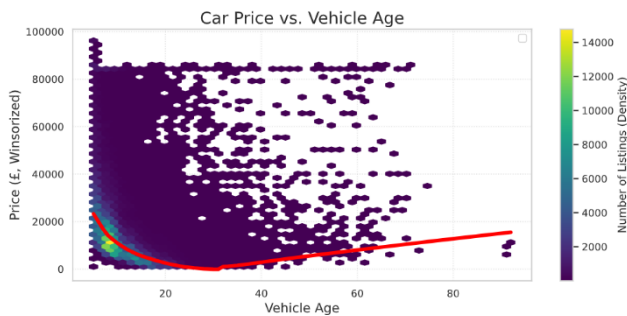


Figure 15: Hexbin density plot showing the relationship between vehicle_age and price.

Price vs Mileage per Year (Observation and Interpretation)

Mileage per year shows how much a car's been driven. According to Autotrader, more use means faster value loss since constant driving adds damage over time (Woodhouse, 2023). To check this effect, we looked at the relationship between mileage_per_year and price_winsorized. Improving the price limits noise from extreme outliers, the data was further filtered to exclude high-mileage-per-year commercial vehicle outliers (>17,500 miles/year) to focus on the core consumer market.

We made a scatter plot using a log scale on price, so dots don't pile up, and trends are easier to see. Figure 16 highlights a clear, strong negative trend. As annual mileage goes up, the price value of the cars tends to decrease, and cars driven less often stay expensive. Instead of adding miles, owners keep values higher by using vehicles less. However, the decline isn't linear. Once vehicles hit 10,000 miles a year, the fall becomes steep. After that, their value clusters into a low range, which proves that heavy usage reduces market value; reselling with a high price would be challenging. Cars with low yearly usage show higher and steadier prices. Using a log scale makes the sharp early drop clearer, showing how the value falls smoothly after that.

In our dataset majority of the vehicles sit within low and mid-yearly usage levels. Also, these cars fall in the mid to high price range, which aligns with Autotrader's market focus on newer cars or barely used cars. Vehicle with high usage from a separate cluster of lower/budget-friendly prices. This pattern shows how yearly mileage really affects value, helping build pricing models that don't change around.

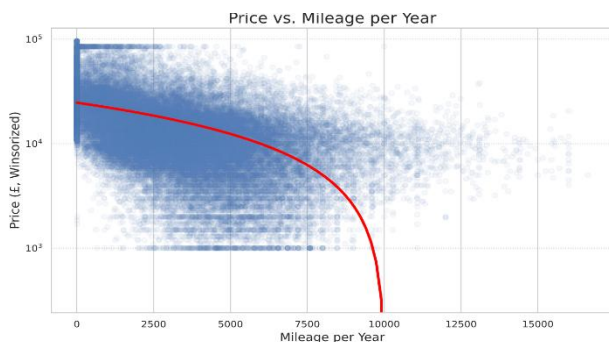


Figure 16: scatter plot showing Price Log Scale vs. Mileage per Year (Engineered Feature).

3.2 Quantitative – Categorical

Price distribution by brand and grouped by vehicle age

The grouped box plot (Figure 17) shows the price distribution for the top 6 volume brands, grouped into four key age brackets: 0-5(new), 6-8(used), 9-15(older), and 16+(vintage). This allows us to calculate brand equity. In general, Figure 17 demonstrates that a car's age is one of the main drivers of price, but its impact depends significantly on brand. Newer cars show the highest price across all the brands, and the price decreases in each older age group. The boxplot is wider for luxury and high-performance brands such as BMW, Audi and Mercedes-Benz in particularly in the newer age groups, which denote larger interquartile ranges in specification and model ranges. In contrast, Nissan, Vauxhall and Volkswagen show narrow ranges along with cheaper price levels, which demonstrate their position in the mainstream market.

For the brand, prices drop as the cars move from one group stage, for example 0-5 age group into the other group (6-8). The decrease is steepest between the 0-5 and 6-8-year groups. The trend becomes flatter after 10 years, suggesting that cars lose most of their value early stages, then stabilise. Vintage cars (16+) show the lowest price for all makes, with a small group of higher value. These higher values denote older models' limit for niche demand rather than standard market pricing.

Figure 17 shows how brand type and age shape price behaviour together. Luxury and high-performance brands hold higher prices in every age group, but the majority remain at lower prices across the full range. The vehicle age creates a consistent descending shift for all brands, while brands difference explains the vertical gap between price levels. This assists in the use of age and brand of cars together when interpreting price variation to customers, other stakeholders and estimating fair market value.

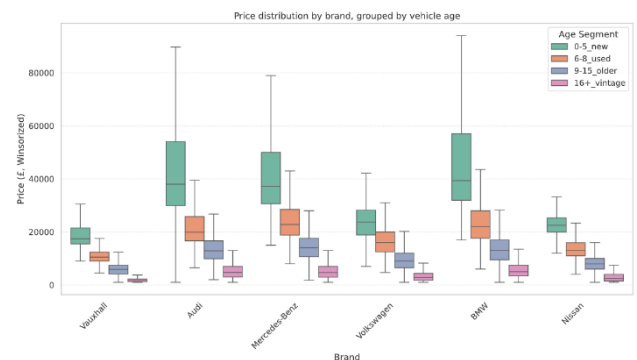


Figure 17: A boxplot showing the price distribution for the top 6 volume brands, grouped into four key age brackets.

Price vs Fuel Type Grouped by Age

Fuel Type	Median Price (0–5 yrs)	Median Price (6–8 yrs)	Retention Ratio	% Drop
Electric	£28,500	£7,507	0.26	74%
Hybrid	£22,490	£9,950	0.44	56%
Diesel	£17,288	£8,900	0.51	49%
Petrol	£13,995	£6,000	0.43	57%

Table 2: Median price, retention, and value change by fuel type

Figure 18 shows the analysis of price and fuel type across age groups, which highlights the difference in value drops. Electric car leads with the highest median price at £28,500, for those under 5 years old, hybrid comes next at £22,490, diesel sits at £17,288 and then petrol at £13,995, indicating that people are buying cars with fuel technologies. However, once cars hit 6-8 years, electric dropped to £7,507, hybrid to £9,950, while diesel and petrol decreased to £8,900 and £6,000.

This pattern denotes that even though EVs begin with a price advantage, they lose more value overall when compared to internal combustion engine (ICE) cars, reducing their value from approximately 53% to 28%. In contrast, hybrid cars keep their values better than electric cars in the used car market, passing them in an average price, whereas diesel cars hold onto 51% of their original value.

Figure 18 shows the trends of how vehicles lose value in uneven ways depending on their fuel. This trend aligns with industry findings that EV depreciation, while historically higher, is becoming more predictable as battery health reporting and technology mature (Cox Automotive, 2025). Even though there aren’t many old electric cars in the data (196), the figure still suggests Autotrader adjust pricing and stock plan as EVs drop fast at first, while petrol and hybrid cars hold better values.

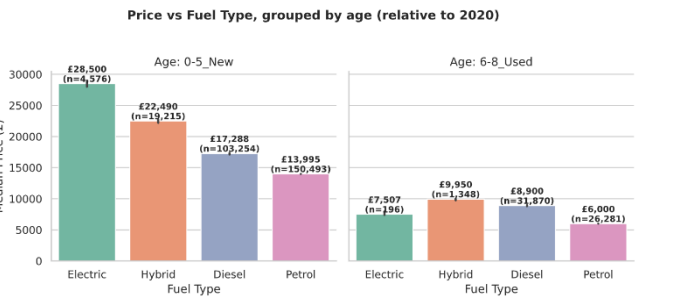


Figure 18: Catplot showing price versus fuel type across age groups.

Mileage & body type

The boxplot shows the analysis of mileage by body type after winsorization to reduce the extreme values. Figure 19 clearly shows the difference in usage intensity among car body types. SUVs and hatchbacks have the lowest median mileage and narrow interquartile ranges, implying these cars are used for short commuting, city journeys. In comparison, Estate pickup cars display the highest median mileages and the widest spread. Suggesting heavier and more unpredictable usage, this could mean it's used for a long journey or a work-related purpose. Saloons, Coupes, Convertibles, and MPVs have mid-range, with moderate median and broader flexibility, suggesting mixed usage trends. The boxplot remains right skewed, with upper whiskers extending further than lower ones, suggesting that very high mileage cars are less common but show in every body type.

Figure 19 denotes that body type is associated with usage intensity, suggesting mileage should be considered jointly with body type, for example, when doing an analysis of customers considering the pricing or condition of the cars.

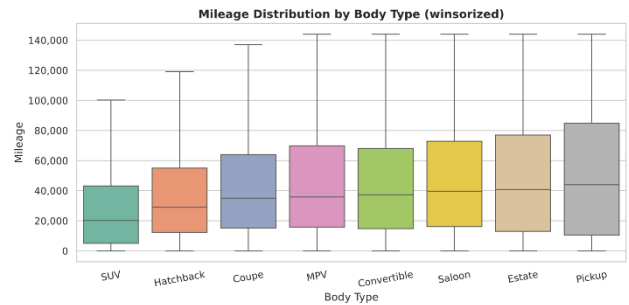


Figure 19: box plot showing the analysis of mileage by body type.

3.3 Categorical - Categorical

Body type and fuel type associations

To find out whether vehicle body type is related to fuel type, we selected the six most common body types and created a visual using row-normalised heatmap. Figure 20 highlights the fuel mix within each body type rather absolute market size. The vehicles that have a sporty and compact style, like convertibles and coupes, use more petrol-powered (76% and 61%), while other body types, such as saloons and estates, heavily use diesel (around 63%). SUVs show a mixed fuel type, with diesel being (51%); however, hybrid is higher than in the smaller body type, suggesting an early electrification trend. Compared to all fuel types of electric cars have insignificant shares, this is because our dataset time frame is up to 2020, when EV adoption was still limited.

A chi-square test is used to confirm the relationship between body type and fuel type. Relationship: ($\chi^2 \approx 52,150.6$, $df = 20$, $p < 0.001$). Figure 20 shows is not random. Patrol fuels are associated with smaller and high-performance cars. Diesel line up with saloon and estate cars that will help with long-distance driving. Hybrids are used more in larger cars. These analyses indicate how different body types are used in the market.

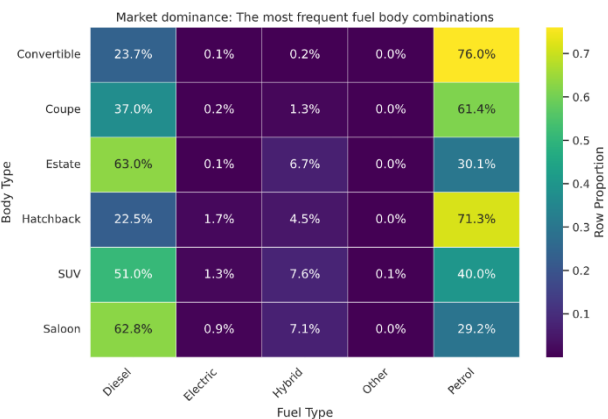


Figure 20: A heatmap showing the fuel type within the 6 most common body types.

Body Type and Price Bucket

Figure 21 shows a strong segmentation of the price bucket by body. Hatchbacks have the highest budget-friendly value at 88%, suggesting their lower cost puts them in good market positioning. Coupes and convertibles have a higher share of mid-range and luxury/high-performance price buckets compared to other body types. For instance, convertibles have

percentages around 60% budget, 29% mid-range, and 11% luxury/high-performance, while coupes rise more in mid-range and luxury mix. Saloons and estates show balanced proportion, maintaining a majority in budget share of 69% and 58% but show a decent amount in mid-range 29% and 38%. The figure also shows that SUVs are the most evenly split with all the price bucket groups, 51% budget, 42% mid-range and 6.5 %luxury, suggesting that customers have a lot of options within the SUV cars, from cheap to expensive.

The bar chart shows a trend that’s reasonable, smaller cars like hatchbacks, most of which are budget-friendly, as they are cheaper and aim to make customers buy them quickly. High-performance and sports cars, such as coupes and convertibles, are more expensive, so they fall into mid-range and luxury group categories. SUVs are in multiple price categories, indicating that a lot of revenue is made from SUV cars and attracts many types of customers.

This analysis proves that body type is a vital factor in pricing. Figure 21 indicates that price distribution can be structured by body type, not randomly.

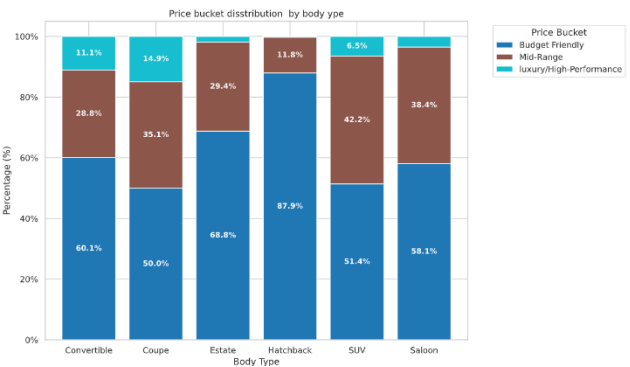


Figure 21: bar chart showing how different body types are spread across price categories.

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